

Montgomery Curves: Exact Enumeration and Probabilistic Analysis

Tsai Yi-Ju

tsaiyiju@sjtu.edu.cn
Shanghai Jiao Tong University

September 1, 2025

Abstract

This paper addresses the question of how many distinct Montgomery curves exist over a finite field $\mathbb{F}_p$ for a given order. We provide a complete answer by presenting precise counting formulas from two perspectives: (1) the number of $\mathbb{F}_p$ -isomorphism classes, and (2) the number of coefficient pairs (A, B) . Additionally, we propose conjectural formulas that approximate the probability of a randomly chosen curve having an order with a specific, cryptographically relevant structure. As a byproduct of our methods, we also establish a novel identity for the Kronecker-Hurwitz class number.

1 Introduction

Since the 1980s, with the advent of Elliptic Curve Cryptography (ECC), their practical value in the field of information security has become increasingly prominent. Among the many models of elliptic curves, the Montgomery curve, defined by its unique algebraic form $By^2 = x^3 + Ax^2 + x$, has garnered significant attention due to its exceptional efficiency in scalar multiplication and its inherent resistance to side-channel attacks[1]. The most famous application is Curve25519, designed by Daniel J. Bernstein[1]. X25519 has been widely deployed, particularly within the TLS 1.3 protocol.

Despite the immense practical success of Montgomery curves, a systematic theoretical investigation of their fundamental mathematical properties over a given finite field remains an area for deeper exploration. There are three fundamental and important questions:

1. **The problem of counting isomorphism classes:** For a prime field $\mathbb{F}_p$ and a given trace t , how many $\mathbb{F}_p$ -isomorphism classes of Montgomery curves of order $p + 1 - t$ exist?

2. **The problem of order distribution:** If the parameters (A, B) are chosen uniformly at random from the set of all valid parameters, what is the probability that the Montgomery curve defined by $By^2 = x^3 + Ax^2 + x$ has a specific order $p + 1 - t$?
3. **The problem of cryptographically secure structures:** If the parameters (A, B) (especially limit that $B = 1$) are chosen uniformly at random from the set of all valid parameters, what is the probability that the Montgomery curve has an order with a specific structure, such as a small cofactor multiplied by a prime (e.g., $4q$ or $8q$)?

The answers to these questions not only deepen our theoretical understanding of this special family of curves but also provide a solid mathematical foundation for securely and efficiently generating and selecting curve parameters in cryptography.

For general elliptic curves represented in the short Weierstrass form ($y^2 = x^3 + ax + b$), the aforementioned questions are well-understood. Precise mathematical formulas exist, particularly for the counting of isomorphism classes and the distribution of orders, which provide clear answers to the first two questions. Specifically, for curves in short Weierstrass form:

1. For Question 1, René Schoof obtained a precise formula in 1987 [7].
2. For Question 2, B. J. Birch provided an exact result in 1968 [2].
3. For Question 3, Steven D. Galbraith and James McKee presented conjectured asymptotic results in 1999 [3].

However, as a specific subset of elliptic curves, Montgomery curves possess properties that differ from the general Weierstrass case due to the constraints of their algebraic form. Consequently, the existing general formulas cannot be directly applied. Although some attention has been paid to Question 2 for Montgomery curves, nice results have not been achieved (see p. 449 of [5]).

This paper provides precise formulas that answer Questions 1 and 2, presented in Theorems 2, 4, and 6. Furthermore, for Question 3, we propose several conjectures (see Conjecture 1, 2, and 3). Although they remain unproven, the conjectures are reasonable. In addition, experimental results indicate that the formulas provided by these conjectures become more precise as p grows. For 64-bit primes, the predictions are accurate to about 2%. This paper also provides a necessary and sufficient condition under which an elliptic curve E can be transformed into a Montgomery curve. We also derive a novel identity involving the Kronecker-Hurwitz class number.

2 Main Results

2.1 Answer for the Question 1

The following two theorems provide a complete answer to Question 1, distinguishing between the cases based on the value of p modulo 4. The definition of

the Kronecker class number is given in [7] as (2.1) Definition.

Theorem 2. Let p be a prime with $p > 3$ and $p \equiv 1 \pmod{4}$. Let $IGM(p, t)$ denote the set of all Montgomery curves over $\mathbb{F}_p$ with order $p + 1 - t$, and let $\sim$ be the equivalence relation of $\mathbb{F}_p$ -isomorphism. Then the cardinality of the quotient set (that is: the number of $\mathbb{F}_p$ -isomorphism classes in $IGM(p, t)$) is given by:

$$|IGM(p, t)/\sim| = \begin{cases} H(t^2 - 4p) & \text{if } t \equiv 2 \pmod{4} \text{ and } |t| < 2\sqrt{p}, \\ 0 & \text{otherwise} \end{cases},$$

where H is the Kronecker class number.

Theorem 4. Let p be a prime such that $p > 3$ and $p \equiv 3 \pmod{4}$. Let $IGM(p, t)$ denote the set of all Montgomery curves over $\mathbb{F}_p$ with order $p + 1 - t$, and let $\sim$ be the equivalence relation of $\mathbb{F}_p$ -isomorphism. Then the number of $\mathbb{F}_p$ -isomorphism classes in $IGM(p, t)$ is given by: $|IGM(p, t)/\sim| =$

$$\begin{cases} H(t^2 - 4p) & \text{if } |t| < 2\sqrt{p} \text{ and } p + 1 - t \equiv 0 \pmod{8} \\ H(t^2 - 4p) - H\left(\frac{t^2 - 4p}{4}\right) & \text{if } |t| < 2\sqrt{p} \text{ and } p + 1 - t \equiv 4 \pmod{8}, \\ 0 & \text{otherwise} \end{cases},$$

where H is the Kronecker class number.

Example 1. Note: The method of counting $\mathbb{F}_p$ -isomorphism classes via their j -invariants is valid for $t \neq 0$. This correspondence breaks down in the case $t = 0$, which involves supersingular curves. We provide the cases where $p + 1 \equiv 0 \pmod{8}$ and $p + 1 \equiv 4 \pmod{8}$.

Table 1: Example for $\mathbb{F}_{29}$.

$p + 1 - t$	t	$D = t^2 - 4p$	$H(D)$	$ IGM(p, t)/\sim $
20	10	-16	2	2
24	6	-80	6	6
28	2	-112	4	4
32	-2	-112	4	4
36	-6	-80	6	6
40	-10	-16	2	2

Table 2: Example for $\mathbb{F}_{31}$.

$p + 1 - t$	t	$D = t^2 - 4p$	$H(D)$	$H(D/4)$	$ IGM(p, t)/\sim $
24	8	-60	4	2	4
28	4	-108	6	2	$6 - 2 = 4$
32	0	-124	6	3	6
36	-4	-108	6	2	$6 - 2 = 4$
40	-8	-60	4	2	4

Table 3: Example for $\mathbb{F}_{43}$.

$p+1-t$	t	$D = t^2 - 4p$	$H(D)$	$H(D/4)$	$ IGM(p, t)/\sim $
32	12	-28	2	1	2
36	8	-108	6	2	$6 - 2 = 4$
40	4	-156	8	4	8
44	0	-172	4	1	$4 - 1 = 3$
48	-4	-156	8	4	8
52	-8	-108	6	2	$6 - 2 = 4$
56	-12	-28	2	1	2

2.2 Answer for the Question 2

The following theorem provide a complete answer to Question 2. The definition of the Kronecker-Hurwitz class number is given in p. 21 of [4].

Theorem 6. Let $p > 3$ be a prime. The cardinality of the set $S_M(p, t) = \{(A, B) \in \mathbb{F}_p^2 \mid |M_{A,B}(\mathbb{F}_p)| = p + 1 - t \text{ and } B(A^2 - 4) \neq 0\}$ is given by the following formula:

$$|S_M(p, t)| = \begin{cases} 3(p-1)H_w\left(\frac{t^2-4p}{4}\right) & \text{if } 4 \mid p+1-t \text{ and } t^2 < 4p, \\ 0 & \text{otherwise} \end{cases},$$

where $M_{A,B}$ denote a Montgomery curve given by $By^2 = x^3 + Ax^2 + x$ and H_w is the Kronecker-Hurwitz class number.

Example 2. The choices for the primes p here are the same as before.

Table 4: Example for $\mathbb{F}_{29}$.

$p+1-t$	t	$D/4 = (t^2 - 4p)/4$	$3(p-1)H_w(D/4)$	$ S_M(p, t) $
20	10	-4	42	42
24	6	-20	168	168
28	2	-28	168	168
32	-2	-28	168	168
36	-6	-20	168	168
40	-10	-4	42	42

Table 5: Example for $\mathbb{F}_{31}$.

$p+1-t$	t	$D/4 = (t^2 - 4p)/4$	$3(p-1)H_w(D/4)$	$ S_M(p, t) $
24	8	-15	180	180
28	4	-27	120	120
32	0	-31	270	270
36	-4	-27	120	120
40	-8	-15	180	180

Table 6: Example for $\mathbb{F}_{43}$.

$p+1-t$	t	$D/4 = (t^2 - 4p)/4$	$3(p-1)H_w(D/4)$	$ S_M(p, t) $
32	12	-7	126	126
36	8	-27	168	168
40	4	-39	504	504
44	0	-43	126	126
48	-4	-39	504	504
52	-8	-27	168	168
56	-12	-7	126	126

2.3 Answer for the Question 3

The vast majority of Montgomery curves in modern applications are of the form $y^2 = x^3 + Ax^2 + x$, which corresponds to setting the parameter $B = 1$. This specific case is of great practical interest for generating suitable curves and also leads to remarkably elegant theoretical results. For these reasons, the conjectures presented below will pertain exclusively to the case where $B = 1$.

These conjectures are motivated by the observation that our exact counting formulas for Montgomery curves are formally analogous to those for general elliptic curves. We therefore hypothesized that the probability distributions would also be similar. By adapting the heuristics from the work of Galbraith and McKee etc. [3], we arrived at the following results.

Conjecture 1. Let $p > 3$ be a prime such that $p \equiv 1 \pmod{4}$. Consider the family of Montgomery curves $E_A : y^2 = x^3 + Ax^2 + x$, where the parameter A is chosen uniformly at random from $\mathbb{F}_p \setminus \{+2, -2\}$. Let $o = |E_A(\mathbb{F}_p)|$ be the order of the curve.

Let $\mathcal{P}_1(p)$ be the probability that the order o is four times a prime (i.e., $o/4$ is prime). Let $\mathcal{P}_2(p)$ be the probability that the order o is eight times a prime (i.e., $8|o$ and $o/8$ is prime).

Then, as $p \rightarrow \infty$, we conjecture that $\mathcal{P}_1(p)$ and $\mathcal{P}_2(p)$ are asymptotically equal, and their value is given by:

$$\mathcal{P}_1(p) \sim \mathcal{P}_2(p) \sim \frac{C_2}{2} \cdot \frac{\pi(p)}{p} \cdot f(p),$$

where $C_2 \approx 0.6601618$ is the Hardy-Littlewood twin prime constant, $\pi(p)$ is the prime-counting function and $f(p)$ is a correction factor dependent on the prime factors of $p-1$, defined as:

$$f(p) = \prod_{\substack{l|(p-1) \\ l \text{ is prime}, l > 2}} \left(1 + \frac{1}{(l-2)(l+1)}\right).$$

Example 3. The primes chosen for this example are the smallest ones satisfying $p \equiv 1 \pmod{4}$ that are greater than 2^{24} , 2^{32} , 2^{48} , and 2^{64} , respectively. This

selection showcases the effect of the factor $f(p)$, as the corresponding values are in some cases close to 1 and in others slightly exceed 1.28. The probabilities were approximated by randomly sampling 200,000 values of A , and are denoted by $\mathcal{P}'_1$ and $\mathcal{P}'_2$.

We define the relative error δ_1 of the conjecture and the relative difference δ_2 of the experimental probabilities as follows:

$$\delta_1 = \frac{\left| \frac{C_2}{2} \frac{\pi(p)}{p} f(p) - \frac{\mathcal{P}'_1 + \mathcal{P}'_2}{2} \right|}{\frac{\mathcal{P}'_1 + \mathcal{P}'_2}{2}}, \quad \delta_2 = \frac{2|\mathcal{P}'_1 - \mathcal{P}'_2|}{\mathcal{P}'_1 + \mathcal{P}'_2}.$$

Table 7: Example for Conjecture 1.

p	$f(p)$	$\pi(p)/p$	$\mathcal{P}'_1(p)$	$\mathcal{P}'_2(p)$	δ_1	δ_2
$2^{24} + 73$	1.00941	0.0642460	0.02193	0.02277	4.224%	3.758%
$2^{32} + 61$	1.00000	0.0473299	0.01582	0.01711	5.115%	7.835%
$2^{48} + 21$	1.28125	0.0310194	0.012965	0.01397	2.591%	7.462%
$2^{64} + 13$	1.02500	0.0230748	0.007945	0.00799	2.014%	0.565%

Conjecture 2. Let $p > 3$ be a prime such that $p \equiv 3 \pmod{4}$. Consider the family of Montgomery curves $E_A : y^2 = x^3 + Ax^2 + x$, where the parameter A is chosen uniformly at random from $\mathbb{F}_p \setminus \{+2, -2\}$. Let $o = |E_A(\mathbb{F}_p)|$ be the order of the curve.

Let $\mathcal{P}_1(p)$ be the probability that the order o is four times a prime (i.e., $o/4$ is prime). Let $\mathcal{P}_2(p)$ be the probability that the order o is eight times a prime (i.e., $8|o$ and $o/8$ is prime).

Then, as $p \rightarrow \infty$, we conjecture that:

$$\mathcal{P}_1(p) \sim \frac{2}{3} \mathcal{P}_2(p),$$

and that $\mathcal{P}_1(p)$ is given by:

$$\mathcal{P}_1(p) \sim \frac{C_2}{2} \cdot \frac{\pi(p)}{p} \cdot f(p).$$

where C_2 is the Hardy-Littlewood twin prime constant, $\pi(p)$ is the prime-counting function, and $f(p)$ is defined as in Conjecture 1.

Example 4. The primes chosen for this example are the smallest ones satisfying $p \equiv 3 \pmod{4}$ that are greater than 2^{24} , 2^{32} , 2^{48} , and 2^{64} , respectively. This selection showcases the effect of the factor $f(p)$, as the corresponding values are in some cases close to 1 and in others slightly exceed 1.33. The probabilities were approximated by randomly sampling 200,000 values of A , and are denoted by $\mathcal{P}'_1$ and $\mathcal{P}'_2$.

We define the relative error δ_1 of the conjecture and the relative difference δ_2 of the experimental probabilities as follows:

$$\delta_1 = \frac{\left| \frac{C_2}{2} \frac{\pi(p)}{p} f(p) - \mathcal{P}'_1 \right|}{\mathcal{P}'_1}, \quad \delta_2 = \frac{2|\mathcal{P}'_1 - \frac{2}{3}\mathcal{P}'_2|}{\mathcal{P}'_1 + \frac{2}{3}\mathcal{P}'_2}.$$

Table 8: Example for Conjecture 1.

p	$f(p)$	$\pi(p)/p$	$\mathcal{P}'_1(p)$	$\mathcal{P}'_2(p)$	δ_1	δ_2
$2^{24} + 43$	1.00207	0.0642460	0.021345	0.03308	0.4428%	3.264%
$2^{32} + 15$	1.31952	0.0473299	0.02106	0.032775	2.116%	3.682%
$2^{48} + 75$	1.33167	0.0310194	0.013725	0.021605	0.6569%	4.823%
$2^{64} + 51$	1.26653	0.0230748	0.009565	0.014875	0.8536%	3.610%

Conjecture 3. Let the prime p and the curve E_A be defined as in Conjecture 2, i.e., $p > 3$ and $p \equiv 3 \pmod{4}$. Let $o = |E_A(\mathbb{F}_p)|$ be the order of the curve.

Let $c = \mathcal{P}_1(p)$ be the probability that $o/4$ is prime.

Then, for a small integer $n \geq 1$, let $\mathcal{P}_n(p)$ be the probability that $4n|o$ and that $o/(4n)$ is a prime number. We conjecture that $\mathcal{P}_n(p)$ is asymptotically related to c as follows:

$$\mathcal{P}_n(p) \sim \begin{cases} c/n & \text{if } n \text{ is odd} \\ 3c/n & \text{if } n \text{ is even} \end{cases}.$$

2.4 Other Results

Proposition 1. 1. For prime $p > 3$ with $p \equiv 1 \pmod{4}$, the following identity holds:

$$\sum_{\substack{t \in \mathbb{Z}, t^2 < p \\ t \equiv 1 \pmod{2}}} H_w(t^2 - p) = \frac{p-2}{3}.$$

2. For prime $p > 3$ with $p \equiv 3 \pmod{4}$, the following identity holds:

$$\sum_{\substack{t \in \mathbb{Z}, t^2 < p \\ t \equiv 0 \pmod{2}}} H_w(t^2 - p) = \frac{p-2}{3}.$$

Proof. The result follows by combining Theorem 6 with the fact that there are a total of $(p-1)(p-2)$ valid pairs of parameters (A, B) . $\square$

Proposition 2. Let E be a non-singular elliptic curve over $\mathbb{F}_p$ (with $p > 3$) whose order is divisible by 4. Then there exists no Montgomery curve over $\mathbb{F}_p$ that is $\mathbb{F}_p$ -isomorphic to E if and only if all three of the following conditions hold:

1. $p \equiv 3 \pmod{4}$.

2. The order of E is not divisible by 8 (i.e., $|E(\mathbb{F}_p)| \equiv 4 \pmod{8}$).
3. The 2-torsion subgroup of E is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$ (i.e., $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$).

Proof. This proposition is a direct consequence of combining Theorem 1 and Theorem 3 from this paper. $\square$

3 Proofs of Theorems

Lemma 1. Let E be a non-singular elliptic curve defined over $\mathbb{F}_p$, where p is prime and $p > 3$. Then there exist positive integers n_1 and n_2 such that the group of points of E is isomorphic to $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$. Further, $n_2 | n_1$ and $n_2 | (p-1)$.

Proof. See Theorem 4.1 in [11], Theorem 6.1 in [9], and Exercise 5.6 in [8]. $\square$

Lemma 2. Let E be a non-singular elliptic curve over the finite field $\mathbb{F}_p$ defined by the equation $y^2 = x^3 + ax^2 + bx + c$, where $p > 3$ is a prime and $\Delta \neq 0$. A point P order is 2 if and only if P can be represented in the form (x, y) and its y -coordinate is 0.

Proof. From Group Law Algorithm 2.3 in Chapter III of [8], we can obtain: if a point P on the elliptic curve E can be represented by (x, y) , then $-P = (x, -y - 0x - 0) = (x, -y)$.

($\Rightarrow$) Suppose the order of P is 2. Then $P \neq \mathcal{O}$ and $2P = P + P = \mathcal{O}$, which implies that P can be represented in the form (x, y) and $P = -P$. For a point $P = (x, y)$, this means $(x, y) = (x, -y)$, so $y = 0$.

($\Leftarrow$) Conversely, suppose a point P is of the form (x, y) with $y = 0$. Then $P \neq \mathcal{O}$ and $P = (x, -0) = -P$. Therefore, $2P = P + (-P) = \mathcal{O}$. Thus, P is a point of order 2. $\square$

Lemma 3. Let M be a non-singular Montgomery curve over the finite field $\mathbb{F}_p$, where $p > 3$ is a prime. Then the order of M is divisible by 4.

Proof. This is a well-known result. See Section 1.3 of [6]. $\square$

Lemma 4. Let E be an elliptic curve over the finite field $\mathbb{F}_p$ defined by the equation $y^2 = x^3 + ax + b$, where $p > 3$ is a prime and $27b^2 + 4a^3 \neq 0$. There exists a Montgomery curve M that is $\mathbb{F}_p$ -isomorphic to E if and only if both of the following conditions hold:

1. The order of E is divisible by 4.
2. There exists an $\alpha \in \mathbb{F}_p$ such that $\alpha^3 + a\alpha + b = 0$ and $3\alpha^2 + a$ is a quadratic residue in $\mathbb{F}_p$.

Proof. See Proposition 1 in [6]. $\square$

Lemma 5. Let $p > 3$ be a prime and a_1, a_2, a_3 be integers. If the congruence equation $x^3 + a_1x^2 + a_2x + a_3 \equiv 0 \pmod{p}$ has three distinct solutions, and p does not divide the discriminant $D = a_1^2a_2^2 - 4a_2^3 - 4a_1^3a_3 - 27a_3^2 + 18a_1a_2a_3$, then the Legendre symbol $\left(\frac{D}{p}\right) = 1$. Furthermore, if $\left(\frac{D}{p}\right) = 1$ and the equation has at least one solution, then it has three distinct solutions.

Proof. See Theorem 1.1 in [10]. $\square$

Lemma 6. Let E be an elliptic curve over the finite field $\mathbb{F}_p$ defined by the equation $v^2 = u^3 + au + b$, where $p > 3$ is a prime and $27b^2 + 4a^3 \neq 0$. If E has a point of order 4, then there exists a Montgomery curve that is $\mathbb{F}_p$ -isomorphic to E .

Proof. Let this point of order 4 be P . Since it is not the point at infinity, it can be represented by (x, y) , and $y \neq 0$. Also, $2P$ is a point of order 2, so let $2P = (\alpha, 0)$ (Lemma 2). Then: $(\alpha, 0) = 2P = 2(x, y)$. The formulas for point doubling give us the following relations: $\alpha = \lambda^2 - 2x$ and $0 = \lambda(x - \alpha) - y$ and $\lambda = \frac{3x^2 + a}{2y}$. This gives: $3\alpha^2 + a = (x - \alpha)^2$. Since there is a point P of order 4, the order of E is a multiple of 4. Furthermore, $(\alpha, 0)$ is on E , so $\alpha^3 + a\alpha + b = 0$, and $3\alpha^2 + a$ is a square in $\mathbb{F}_p$. Therefore, by Lemma 4, there exists a Montgomery curve that is $\mathbb{F}_p$ -isomorphic to E .

Note: $\alpha = \lambda^2 - 2x$ and $0 = \lambda(x - \alpha) - y$ and $\lambda = \frac{3x^2 + a}{2y} \Rightarrow \alpha = \left(\frac{3x^2 + a}{2y}\right)^2 - 2x$ and $2y^2 = (3x^2 + a)(x - \alpha) \Rightarrow \alpha + 2x = \frac{3x^2 + a}{2(x - \alpha)} \Rightarrow 2(\alpha + 2x)(x - \alpha) = 3x^2 + a \Rightarrow -2\alpha^2 + 4x^2 - 2\alpha x = 3x^2 + a \Rightarrow x^2 - 2\alpha x = 2\alpha^2 + a \Rightarrow x^2 - 2\alpha x + \alpha^2 = 3\alpha^2 + a$. $\square$

Lemma 7. Let $p > 3$ be a prime. Let $IG(p, t)$ denote the set of all elliptic curves over $\mathbb{F}_p$ with order $p + 1 - t$, and let $\sim$ be the equivalence relation of $\mathbb{F}_p$ -isomorphism. Then the cardinality of the quotient set (that is: the number of $\mathbb{F}_p$ -isomorphism classes in $IG(p, t)$) satisfies:

$$|IG(p, t)/\sim| = \begin{cases} H(t^2 - 4p) & \text{if } |t| \leq 2\sqrt{p} \\ 0 & \text{if } |t| > 2\sqrt{p} \end{cases},$$

where H is the Kronecker class number, not the Kronecker-Hurwitz class number, the latter of which, in this paper, is denoted by H_w .

Proof. See (4.6) Theorem in [7]. $\square$

Lemma 8. Let $p > 3$ be a prime. Let $IGT(p, t)$ denote the set of all elliptic curves E over $\mathbb{F}_p$ with order $p + 1 - t$ and $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Let $\sim$ be the equivalence relation of $\mathbb{F}_p$ -isomorphism. Then the number of $\mathbb{F}_p$ -isomorphism classes in $IGT(p, t)$ is given by:

$$|IGT(p, t)/\sim| = \begin{cases} H\left(\frac{t^2 - 4p}{4}\right) & \text{if } t^2 < 4p \text{ and } 4 \mid p + 1 - t \\ 0 & \text{otherwise} \end{cases},$$

where H is the Kronecker class number.

Proof. See (4.9) Theorem in [7]. In the case where $p \equiv 3 \pmod{4}$ and $t = 0$, we refer to (4.8) Lemma in [7]. Furthermore, since p is a prime number, it follows that $H(-p) = h(-p)$. $\square$

Lemma 9. Let M be a non-singular Montgomery curve over the finite field $\mathbb{F}_p$, where p is a prime with $p > 3$ and $p \equiv 3 \pmod{4}$. If the 2-torsion subgroup $M(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, then the order of M is divisible by 8.

Proof. Let the Montgomery curve M be defined by the equation $By^2 = x^3 + Ax^2 + x$ ($B(A^2 - 4) \neq 0$). Since the 2-torsion subgroup $M(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, M has three distinct points of order 2. According to Proposition 3 in [6], this condition is equivalent to $A^2 - 4$ being a non-zero quadratic residue in $\mathbb{F}_p$. This means the Legendre symbol $\left(\frac{A^2-4}{p}\right) = 1$, which implies $\left(\frac{A-2}{p}\right)\left(\frac{A+2}{p}\right) = 1$. This holds if and only if $(A-2)$ and $(A+2)$ are either simultaneously quadratic residues or simultaneously quadratic non-residues in $\mathbb{F}_p$. And $p \equiv 3 \pmod{4}$. By Theorem 1 in the same reference [6], the order of the curve is divisible by 8. $\square$

Theorem 1. Let E be a non-singular elliptic curve defined over $\mathbb{F}_p$, where p is a prime with $p > 3$ and $p \equiv 1 \pmod{4}$. If the order of E is a multiple of 4, then there exists a Montgomery curve that is $\mathbb{F}_p$ -isomorphic to E .

Proof. Over a field K with characteristic not equal to 2 or 3, any elliptic curve is K -isomorphic to a curve in short Weierstrass form (See III.1 in [8]). Let E be defined by the equation $y^2 = x^3 + ax + b$ with $27b^2 + 4a^3 \neq 0$. By Lemma 1, the group of points of E is isomorphic to $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$, with $n_2 \mid n_1$. Let $n_1 = 2^{k_1}r_1$ and $n_2 = 2^{k_2}r_2$, where $k_1, k_2 \in \mathbb{Z}, k_1, k_2 \geq 0$, and r_1, r_2 are odd. Since $4 \mid |E(\mathbb{F}_p)| = n_1n_2$, it follows that $k_1 + k_2 \geq 2$. As $n_2 \mid n_1$, we have $k_1 \geq k_2 \geq 0$.

1. When $k_2 = 0$, it follows that $k_1 \geq 2$. Thus, $4 \mid n_1$, and $\mathbb{Z}_{n_1}$ has an element of order 4, namely $n_1/4$. Therefore, $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$ has an element of order 4, namely $(n_1/4, 0)$. This means E has a point of order 4. By Lemma 6, E is $\mathbb{F}_p$ -isomorphic to a Montgomery curve.
2. When $k_2 \geq 1$, we have $k_1 \geq k_2 \geq 1$. This implies that $2 \mid n_1$ and $2 \mid n_2$. Thus, $\mathbb{Z}_{n_1}$ and $\mathbb{Z}_{n_2}$ each have an element of order 2, namely $n_1/2$ and $n_2/2$ respectively. Therefore, $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$ has 3 distinct elements of order 2: $(n_1/2, 0)$, $(n_1/2, n_2/2)$, and $(0, n_2/2)$. So, E has 3 distinct points of order 2. By Lemma 2, let them be $(\alpha_1, 0)$, $(\alpha_2, 0)$, $(\alpha_3, 0)$, where $\alpha_1, \alpha_2, \alpha_3$ are distinct. Since these points are on E , they are all solutions to the equation $z^3 + az + b = 0$.

A derivation using Vieta's formulas shows that $(3\alpha_1^2 + a)(3\alpha_2^2 + a)(3\alpha_3^2 + a) = 27(\alpha_1^2\alpha_2^2\alpha_3^2) + 9a(\alpha_1^2\alpha_2^2 + \alpha_2^2\alpha_3^2 + \alpha_3^2\alpha_1^2) + 3a^2(\alpha_1^2 + \alpha_2^2 + \alpha_3^2) + a^3 = 27(\alpha_1\alpha_2\alpha_3)^2 + 9a((\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3)^2 - 2\alpha_1\alpha_2\alpha_3(\alpha_1 + \alpha_2 + \alpha_3)) + 3a^2((\alpha_1 + \alpha_2 + \alpha_3)^2 - 2(\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3)) + a^3 = 27(-b)^2 + 9a(a^2 - 2(-b)(0)) + 3a^2(0^2 - 2a) + a^3 = 27b^2 + 4a^3$.

Furthermore, since the equation $z^3 + az + b = 0$ has three distinct solutions, by Lemma 5, $D = -(4a^3 + 27b^2)$ is a quadratic residue in $\mathbb{F}_p$. Also, since $p \equiv 1 \pmod{4}$, -1 is a quadratic residue. Therefore, the product $(3\alpha_1^2 + a)(3\alpha_2^2 + a)(3\alpha_3^2 + a)$ is a quadratic residue. This implies that $\left(\frac{3\alpha_1^2 + a}{p}\right) \left(\frac{3\alpha_2^2 + a}{p}\right) \left(\frac{3\alpha_3^2 + a}{p}\right) = 1$.

Thus, at least one of $(3\alpha_1^2 + a)$, $(3\alpha_2^2 + a)$, and $(3\alpha_3^2 + a)$ is a quadratic residue. Assuming it is $(3\alpha_i^2 + a)$, and because $\alpha_i^3 + a\alpha_i + b = 0$, it follows from Lemma 4 that E is $\mathbb{F}_p$ -isomorphic to a Montgomery curve.

□

Theorem 2. Let p be a prime with $p > 3$ and $p \equiv 1 \pmod{4}$. Let $IGM(p, t)$ denote the set of all Montgomery curves over $\mathbb{F}_p$ with order $p + 1 - t$, and let $\sim$ be the equivalence relation of $\mathbb{F}_p$ -isomorphism. Then the cardinality of the quotient set (that is: the number of $\mathbb{F}_p$ -isomorphism classes in $IGM(p, t)$) is given by:

$$|IGM(p, t) / \sim| = \begin{cases} H(t^2 - 4p) & \text{if } t \equiv 2 \pmod{4} \text{ and } |t| < 2\sqrt{p}, \\ 0 & \text{otherwise} \end{cases},$$

where H is the Kronecker class number.

Proof. Let $N(p, t)$ denote $|IGM(p, t) / \sim|$, where $\sim$ is $\mathbb{F}_p$ -isomorphism.

1. When $|t| > 2\sqrt{p}$, $IGM(p, t)$ is the empty set. When $4 \nmid (p + 1 - t)$, by Lemma 3, $IGM(p, t)$ is the empty set. Therefore, when $|t| > 2\sqrt{p}$ or $t \not\equiv 2 \pmod{4}$, we have $N(p, t) = 0$.

2. When $|t| < 2\sqrt{p}$ and $t \equiv 2 \pmod{4}$. By Lemma 7, let $IG(p, t) / \sim = \{[E_1], \dots, [E_{H(t^2 - 4p)}]\}$, where $IG(p, t)$ is the set of all elliptic curves over $\mathbb{F}_p$ with order $p + 1 - t$ and $[E_i]$ is the class containing the curve E_i .

Since $p \equiv 1 \pmod{4}$ and $4 \mid (p + 1 - t)$, by Theorem 1, for each elliptic curve E_i , there exists a Montgomery curve M_i such that $M_i \sim E_i$.

Furthermore, since $E_i \not\sim E_j$ for $i \neq j$, it follows that $M_i \not\sim M_j$ for $i \neq j$.

Additionally, for any Montgomery curve $M \in IGM(p, t)$, it is also an elliptic curve in $IG(p, t)$. Therefore, it belongs to a unique isomorphism class $[E_r]$ for some r , which implies $M \sim M_r$.

Therefore, when $|t| < 2\sqrt{p}$ and $t \equiv 2 \pmod{4}$, $N(p, t) = H(t^2 - 4p)$.

□

Theorem 3. Let E be an elliptic curve over $\mathbb{F}_p$ with order a multiple of 4, where p is a prime with $p > 3$ and $p \equiv 3 \pmod{4}$. Then E is not $\mathbb{F}_p$ -isomorphic to any Montgomery curve if and only if $|E(\mathbb{F}_p)| \equiv 4 \pmod{8}$ and $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

Proof. ($\Rightarrow$) The proof will be done by proving its contrapositive. I.e.: If $|E(\mathbb{F}_p)| \not\equiv 4 \pmod{8}$ or $E(\mathbb{F}_p)[2] \not\cong \mathbb{Z}_2 \times \mathbb{Z}_2$, then E is $\mathbb{F}_p$ -isomorphic to some Montgomery curve.

Over a field K with characteristic not equal to 2 or 3, any elliptic curve is isomorphic to a curve in short Weierstrass form (See III.1 in [8]). Let E be defined by the equation $y^2 = x^3 + ax + b$ with $4a^3 + 27b^2 \neq 0$.

1. If $|E(\mathbb{F}_p)| \not\equiv 4 \pmod{8}$. Since $|E(\mathbb{F}_p)|$ is a multiple of 4, this implies $|E(\mathbb{F}_p)| \equiv 0 \pmod{8}$. By Lemma 1, let the group of points of E be isomorphic to $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$, where $n_1 = 2^{k_1}r_1$ and $n_2 = 2^{k_2}r_2$ with $2 \nmid r_1, r_2$ and $k_1, k_2 \geq 0$. Since $|E(\mathbb{F}_p)| = n_1n_2 \equiv 0 \pmod{8}$, it follows that $k_1 + k_2 \geq 3$. As $n_2 \mid n_1$, we also have $k_1 \geq k_2 \geq 0$.

If $0 \leq k_2 \leq 1$, then $k_1 \geq 2$. Therefore, $4 \mid n_1$. This implies that $\mathbb{Z}_{n_1}$ has an element of order 4, namely $n_1/4$. Thus, $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$ has an element of order 4, $(n_1/4, 0)$.

If $k_2 \geq 2$, then $4 \mid n_2$. This implies that $\mathbb{Z}_{n_2}$ has an element of order 4, namely $n_2/4$. Thus, $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$ has an element of order 4, $(0, n_2/4)$.

In both cases, E has a point of order 4. By Lemma 6, E is $\mathbb{F}_p$ -isomorphic to some Montgomery curve.

2. If $E[2] \not\cong \mathbb{Z}_2 \times \mathbb{Z}_2$. This means that E does not have three distinct points of order 2. Since $|E|$ is a multiple of 4, E has at least one element of order 2. Therefore, the 2-torsion subgroup of E , $E(\mathbb{F}_p)[2]$, is isomorphic to $\mathbb{Z}_2$.

The order of E is a multiple of 4, but its 2-torsion subgroup has only two elements. This implies, in its group structure, that E contains a cyclic subgroup of order 4. So, E has a point of order 4. By Lemma 6, E is $\mathbb{F}_p$ -isomorphic to some Montgomery curve.

($\Leftarrow$) We prove this by contradiction. Assume that an elliptic curve E satisfies $|E(\mathbb{F}_p)| \equiv 4 \pmod{8}$ and $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, but is $\mathbb{F}_p$ -isomorphic to a Montgomery curve M . Then, $8 \nmid |M(\mathbb{F}_p)|$, and its 2-torsion subgroup $M(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. However, since $p \equiv 3 \pmod{4}$, and $M(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, it follows from Lemma 9 that the order of M is divisible by 8, i.e., $8 \mid |M(\mathbb{F}_p)|$. Therefore, no such Montgomery curve exists under these conditions. $\square$

Theorem 4. Let p be a prime such that $p > 3$ and $p \equiv 3 \pmod{4}$. Let $IGM(p, t)$ denote the set of all Montgomery curves over $\mathbb{F}_p$ with order $p+1-t$, and let $\sim$ be the equivalence relation of $\mathbb{F}_p$ -isomorphism. Then the number of $\mathbb{F}_p$ -isomorphism classes in $IGM(p, t)$ is given by:

$$|IGM(p, t) / \sim| = \begin{cases} H(t^2 - 4p) & \text{if } |t| < 2\sqrt{p} \text{ and } p+1-t \equiv 0 \pmod{8} \\ H(t^2 - 4p) - H\left(\frac{t^2-4p}{4}\right) & \text{if } |t| < 2\sqrt{p} \text{ and } p+1-t \equiv 4 \pmod{8}, \\ 0 & \text{otherwise} \end{cases}$$

where H is the Kronecker class number.

Proof. Let $N(p, t)$ denote $|IGM(p, t)/\sim|$, where $\sim$ is the $\mathbb{F}_p$ -isomorphism relation.

1. When $|t| > 2\sqrt{p}$, $IGM(p, t)$ is the empty set. By Lemma 3, the order of any Montgomery curve is a multiple of 4. Therefore, if $4 \nmid (p+1-t)$, $IGM(p, t)$ is also the empty set. In these cases, we have $N(p, t) = 0$.
2. When $|t| < 2\sqrt{p}$ and $4 \mid (p+1-t)$, we consider two cases:

Case A: $p+1-t \equiv 0 \pmod{8}$

By Lemma 7, the set of isomorphism classes of all elliptic curves with order $p+1-t$ can be written as $IG(p, t)/\sim = \{[E_1], \dots, [E_{H(t^2-4p)}]\}$. For any elliptic curve E_i in this set, its order is $|E_i(\mathbb{F}_p)| = p+1-t \equiv 0 \pmod{8}$. By Theorem 3, for each i there exists a Montgomery curve M_i such that $M_i \sim E_i$.

Furthermore, since $E_i \not\sim E_j$ for $i \neq j$, it follows that $M_i \not\sim M_j$. Moreover, for any Montgomery curve $M \in IGM(p, t)$, it belongs to a unique isomorphism class $[E_r]$, which implies $M \sim E_r \sim M_r$.

Therefore, $N(p, t) = H(t^2 - 4p)$.

Case B: $p+1-t \equiv 4 \pmod{8}$

Let $S = IG(p, t)/\sim$ be the set of all isomorphism classes of elliptic curves with order $p+1-t$, with cardinality $|S| = H(t^2 - 4p)$. Let $B \subset S$ be the subset of classes $[E]$ for which $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Its cardinality is $|B| = H\left(\frac{t^2-4p}{4}\right)$ by Lemma 8. By Theorem 3, none of the classes in B can be $\mathbb{F}_p$ -isomorphic to a Montgomery curve.

We consider the complement of B in S , which is $A = S \setminus B$. The cardinality of this set is $|A| = |S| - |B| = H(t^2 - 4p) - H\left(\frac{t^2-4p}{4}\right)$.

Let $A = \{[E_1], \dots, [E_k]\}$, where $k = H(t^2 - 4p) - H\left(\frac{t^2-4p}{4}\right)$.

For each class $[E_i]$ ($i = 1, \dots, k$) in the set A , by definition of A , the curve E_i satisfies $|E_i(\mathbb{F}_p)| \equiv 4 \pmod{8}$ and $E_i(\mathbb{F}_p)[2] \not\cong \mathbb{Z}_2 \times \mathbb{Z}_2$. By Theorem 3, for each i , there exists a Montgomery curve M_i such that $M_i \sim E_i$.

Furthermore, since $E_i \not\sim E_j$ for $i \neq j$, it follows that $M_i \not\sim M_j$.

Moreover, for any Montgomery curve $M \in IGM(p, t)$, it belongs to some class $[E_r]$ in S . As shown previously, $[E_r]$ cannot be in B , so $[E_r] \in A$. This implies $M \sim E_r$, and since $E_r \sim M_r$, it follows that $M \sim M_r$.

Therefore, $N(p, t) = k = H(t^2 - 4p) - H\left(\frac{t^2-4p}{4}\right)$.

Combining all cases yields the conclusion of the theorem. $\square$

Lemma 10. Let p be a prime such that $p > 3$. The number of solutions to the quadratic congruence

$$x^2 + ax + b \equiv 0 \pmod{p}$$

is determined by its discriminant $D = a^2 - 4b$ and the Legendre symbol $\left(\frac{D}{p}\right)$ as follows:

1. The congruence has no solution if and only if $\left(\frac{D}{p}\right) = -1$.
2. The congruence has two distinct solutions if and only if $\left(\frac{D}{p}\right) = 1$.
3. The congruence has one unique solution if and only if $\left(\frac{D}{p}\right) = 0$.

Proof. Note that $x^2 + ax + b \equiv (x + \frac{a}{2})^2 - (\frac{a^2}{4} - b) \pmod{p}$ and $4 \equiv 2^2 \pmod{p}$. $\square$

Lemma 11. Let E_1 and E_2 be two non-singular elliptic curves defined over the finite field $\mathbb{F}_p$, where $p > 3$ is a prime. Let the equations for the curves be:

$$\begin{aligned} E_1 : \quad y^2 &= x^3 + Ax^2 + Bx, \\ E_2 : \quad Y^2 &= X^3 - 2AX^2 + (A^2 - 4B)X, \end{aligned}$$

where $A, B \in \mathbb{F}_p$ are constants such that the curves are non-singular. Then the orders of E_1 and E_2 , denoted $|E_1(\mathbb{F}_p)|$ and $|E_2(\mathbb{F}_p)|$, are equal.

Proof. The order of a non-singular elliptic curve E defined by the equation $y^2 = f(x)$ over $\mathbb{F}_p$, denoted $|E(\mathbb{F}_p)|$, is given by the formula (See Application 1.3 in Chapter V of [8].):

$$|E(\mathbb{F}_p)| = p + 1 + \sum_{k \in \mathbb{F}_p} \left(\frac{f(k)}{p} \right),$$

where $\left(\frac{\cdot}{p}\right)$ is the Legendre symbol. To prove the lemma, it is sufficient to show that the sums of the Legendre symbols corresponding to E_1 and E_2 are equal.

Let $f_1(x) = x^3 + Ax^2 + Bx$, $f_2(X) = X^3 - 2AX^2 + (A^2 - 4B)X$,

$$S_1 = \sum_{x \in \mathbb{F}_p} \left(\frac{f_1(x)}{p} \right), \text{ and } S_2 = \sum_{X \in \mathbb{F}_p} \left(\frac{f_2(X)}{p} \right).$$

Consider the rational map $X(x) : \mathbb{F}_p^* = \mathbb{F}_p \setminus \{0\} \rightarrow \mathbb{F}_p$ given by:

$$X(x) = x + A + \frac{B}{x}.$$

Therefore,

$$f_1(x) = X(x) \cdot x^2.$$

We now substitute this into the sum S_1 . Note that $f_1(0) = 0$, so $\left(\frac{f_1(0)}{p}\right) = 0$. Thus,

$$S_1 = 0 + \sum_{x \in \mathbb{F}_p^*} \left(\frac{f_1(x)}{p} \right) = \sum_{x \in \mathbb{F}_p^*} \left(\frac{X(x) \cdot x^2}{p} \right) = \sum_{x \in \mathbb{F}_p^*} \left(\frac{X(x)}{p} \right) \left(\frac{x^2}{p} \right) = \sum_{x \in \mathbb{F}_p^*} \left(\frac{X(x)}{p} \right).$$

This sum is indexed by x , but the summand depends on X . We can re-index the sum to be over $X \in \mathbb{F}_p$. To do this, we group terms by the value of $X(x)$.

$$S_1 = \sum_{X \in \mathbb{F}_p} \left(\frac{X}{p} \right) \cdot (\text{number of } x \in \mathbb{F}_p^* \text{ that map to this } X).$$

The number of preimages x for each X is the number of solutions to the quadratic equation $x^2 - (X - A)x + B = 0$. This can be expressed as $1 + \left(\frac{D_X}{p} \right)$, where D_X is the discriminant $D_X = (X - A)^2 - 4B$ (Lemma 10).

The sum S_1 can now be written as:

$$S_1 = \sum_{X \in \mathbb{F}_p} \left(\frac{X}{p} \right) \left(1 + \left(\frac{(X - A)^2 - 4B}{p} \right) \right).$$

Expanding this expression gives two separate sums:

$$S_1 = \sum_{X \in \mathbb{F}_p} \left(\frac{X}{p} \right) + \sum_{X \in \mathbb{F}_p} \left(\frac{X}{p} \right) \left(\frac{(X - A)^2 - 4B}{p} \right).$$

For prime $p > 3$: $\sum_{X \in \mathbb{F}_p} \left(\frac{X}{p} \right) = 0 + \left(\frac{p-1}{2} \right) + (-1)\left(\frac{p-1}{2} \right) = 0$. Using the multiplicative property of the Legendre symbol for the second term, we obtain:

$$S_1 = \sum_{X \in \mathbb{F}_p} \left(\frac{X \cdot ((X - A)^2 - 4B)}{p} \right).$$

Now, re-writing the polynomial $f_2(X)$:

$$f_2(X) = X(X^2 - 2AX + (A^2 - 4B)) = X((X - A)^2 - 4B).$$

Therefore,

$$S_2 = \sum_{X \in \mathbb{F}_p} \left(\frac{X((X - A)^2 - 4B)}{p} \right).$$

By direct comparison, we have shown that $S_1 = S_2$.

Thus, $|E_1(\mathbb{F}_p)| = |E_2(\mathbb{F}_p)|$ □

Lemma 12. Let p be a prime such that $p > 3$. Suppose the two congruences $z^3 + az + b \equiv 0 \pmod{p}$ and $z^3 + a'z + b' \equiv 0 \pmod{p}$ both have three distinct roots in $\mathbb{Z}_p$.

If the coefficient pairs (a, b) and (a', b') are distinct, then the two congruences have at most one common root. Furthermore, if a common root exists, then $a \not\equiv a' \pmod{p}$.

Proof. If there are two common roots $r_1 \not\equiv r_2 \pmod{p}$, then $a \equiv a' \pmod{p}$ and $b \equiv b' \pmod{p}$. If there is a common root and $a \equiv a' \pmod{p}$, then $b \equiv b' \pmod{p}$. □

Lemma 13. Let E be a non-singular elliptic curve defined over a prime field $\mathbb{F}_p$ with $p > 3$, given by the equation $y^2 = x^3 + ax + b$. If the 2-torsion subgroup $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, then the following statements hold:

1. The polynomial $f(x) = x^3 + ax + b$ has three distinct roots r_1, r_2, r_3 in $\mathbb{F}_p$.
2. For each root $r_i \in \{r_1, r_2, r_3\}$, the element $-3r_i^2 - 4a$ is a non-zero quadratic residue in $\mathbb{F}_p$.
3. Let s_i be the smallest integer in $\{0, 1, \dots, p-1\}$ such that $s_i^2 \equiv -3r_i^2 - 4a \pmod{p}$,

$$A_i = \frac{-6r_i}{s_i} \quad \text{and} \quad B_i = \frac{1}{s_i^3}.$$

This construction yields six mutually distinct pairs of coefficients, (A_i, B_i) and $(-A_i, -B_i)$ for $i \in \{1, 2, 3\}$. The six corresponding Montgomery curves, given by the equation

$$By^2 = x^3 + Ax^2 + x,$$

all have the same order as the original curve E .

Proof. 1. The condition $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ implies that E has three distinct points of order two. By Lemma 2, these are of the form $(r_1, 0), (r_2, 0), (r_3, 0)$ for distinct $r_1, r_2, r_3 \in \mathbb{F}_p$. Since these points are on the curve, r_1, r_2, r_3 are the roots.

2. Choose an arbitrary root of $x^3 + ax + b = 0$, say r_i . We define a map with the substitution $x' = x - r_i$ (which implies $x = x' + r_i$) and $y' = y$. This transformation yields the curve

$$E' : \quad y'^2 = x'^3 + (3r_i)x'^2 + (3r_i^2 + a)x'.$$

Since the map is an isomorphism defined over $\mathbb{F}_p$, the curves E and E' are isomorphic over $\mathbb{F}_p$. It follows that $E'(\mathbb{F}_p)[2]$ is also isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Therefore, E' also has three distinct points of order two. This implies that $x'^3 + (3r_i)x'^2 + (3r_i^2 + a)x' = x'(x'^2 + (3r_i)x' + (3r_i^2 + a)) = 0$ has three distinct roots. This implies that the quadratic factor, $x'^2 + (3r_i)x' + (3r_i^2 + a)$, has two distinct roots. By Lemma 10, its discriminant, $D = (3r_i)^2 - 4(3r_i^2 + a) = -3r_i^2 - 4a$, is a non-zero quadratic residue in $\mathbb{F}_p$.

3. By Lemma 11, the curve E'' , given by the equation

$$E'' : \quad y''^2 = x''^3 - 6r_ix''^2 + (-3r_i^2 - 4a)x'',$$

has the same order as the curve E' .

We now show that E'' is $\mathbb{F}_p$ -isomorphic to two Montgomery curves M, M' over $\mathbb{F}_p$. Let s_i be the smallest integer in $\{0, 1, \dots, p-1\}$ such that $s_i^2 \equiv -3r_i^2 - 4a \pmod{p}$. The corresponding Montgomery curves is given by the equations:

$$M : \quad B_i Y^2 = X^3 + A_i X^2 + X, \quad \text{and} \quad M' : \quad -B_i Y'^2 = X'^3 - A_i X'^2 + X',$$

where $A_i = -6r_i/s_i$ and $B_i = 1/s_i^3$.

The $\mathbb{F}_p$ -isomorphism from E'' to M is given by the transformation that maps a point (x'', y'') on E'' to a point (X, Y) on M via:

$$(X, Y) = \left(\frac{x''}{s_i}, y'' \right).$$

To verify this, we substitute the transformation into the equation for M . The right-hand side (RHS) becomes:

$$\begin{aligned} \text{RHS} &= X^3 + A_i X^2 + X = \left(\frac{x''}{s_i} \right)^3 + \left(\frac{-6r_i}{s_i} \right) \left(\frac{x''}{s_i} \right)^2 + \left(\frac{x''}{s_i} \right) \\ &= \frac{x''^3 - 6r_i x''^2 + s_i^2 x''}{s_i^3} \\ &= \frac{y''^2}{s_i^3} \quad (\text{since } (x'', y'') \text{ is on } E''). \end{aligned}$$

The left-hand side (LHS) is $B_i Y^2 = \left(\frac{1}{s_i^3} \right) y''^2$. Since LHS = RHS, the mapping is a valid $\mathbb{F}_p$ -isomorphism. Additionally, M is $\mathbb{F}_p$ -isomorphic to M' via the transformation $(X', Y') = (-X, Y)$. As E'' and M are $\mathbb{F}_p$ -isomorphic, they have the same order. From the expressions for A_i and B_i , it is clear that when $r_i \neq r_j$, at least one of the conditions $A_i \neq A_j$ or $B_i \neq B_j$ holds. Thus, the six pairs are pairwise distinct. $\square$

Lemma 14. Let M be a non-singular Montgomery curve defined over a prime field $\mathbb{F}_p$ with $p > 3$, given by the equation $By^2 = x^3 + Ax^2 + x$, $B(A^2 - 4) \neq 0$. Then the short Weierstrass curve $E : w^2 = z^3 + az + b$, with coefficients defined by

$$a = -\frac{A^2 + 12}{3B^2} \quad \text{and} \quad b = \frac{2A(A^2 - 36)}{27B^3}$$

has the same order with M , and its 2-torsion subgroup $E(\mathbb{F}_p)[2]$ is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Proof. 1. $|M(\mathbb{F}_p)| = |E(\mathbb{F}_p)|$.

Let M be a non-singular Montgomery curve, $M : By^2 = x^3 + Ax^2 + x$, $B(A^2 - 4) \neq 0$. We apply $(X, Y) = (x/B, y/B)$ to transform it into:

$$M' : Y^2 = X^3 + \frac{A}{B}X^2 + \frac{1}{B^2}X.$$

As M and M_1 are $\mathbb{F}_p$ -isomorphic, they have the same order.

By Lemma 11, M'' has the same order with M' . And,

$$M'' : Y'^2 = X'^3 - \frac{2A}{B}X'^2 + \frac{A^2 - 4}{B^2}X'.$$

The curve M'' is in long Weierstrass form, with $c_2 = -2A/B$, $c_4 = (A^2 - 4)/B^2$, and $c_6 = 0$. We can transform it to a short Weierstrass curve $E : w^2 = z^3 + az + b$ by applying a coordinate shift that eliminates the X'^2 term.

The required transformation is $X' = z - c_2/3$ and $Y' = w$. We substitute this into the long Weierstrass equation:

$$w^2 = \left(z - \frac{c_2}{3}\right)^3 + c_2 \left(z - \frac{c_2}{3}\right)^2 + c_4 \left(z - \frac{c_2}{3}\right) + c_6.$$

Expanding the terms on the right-hand side:

$$\begin{aligned} w^2 &= \left(z^3 - c_2 z^2 + \frac{c_2^2}{3} z - \frac{c_2^3}{27}\right) \\ &\quad + \left(c_2 z^2 - \frac{2c_2^2}{3} z + \frac{c_2^3}{9}\right) \\ &\quad + \left(c_4 z - \frac{c_2 c_4}{3}\right) \\ &\quad + c_6. \end{aligned}$$

Combining terms by powers of z , the z^2 term cancels out, and we are left with:

$$w^2 = z^3 + \left(c_4 - \frac{c_2^2}{3}\right) z + \left(\frac{2c_2^3}{27} - \frac{c_2 c_4}{3} + c_6\right).$$

This is a short Weierstrass curve $w^2 = z^3 + az + b$ where $a = c_4 - c_2^2/3$ and $b = 2c_2^3/27 - c_2 c_4/3 + c_6$.

Now, we substitute the specific coefficients of M'' ($c_2 = -2A/B$, $c_4 = (A^2 - 4)/B^2$, $c_6 = 0$):

$$\begin{aligned} a &= \frac{A^2 - 4}{B^2} - \frac{(-2A/B)^2}{3} = \frac{A^2 - 4}{B^2} - \frac{4A^2}{3B^2} = -\frac{A^2 + 12}{3B^2}. \\ b &= \frac{2(-2A/B)^3}{27} - \frac{(-2A/B)((A^2 - 4)/B^2)}{3} = \frac{2A(A^2 - 36)}{27B^3}. \end{aligned}$$

These are precisely the coefficients of the curve E in the lemma. Thus, M'' and E are $\mathbb{F}_p$ -isomorphic, so $|M''(\mathbb{F}_p)| = |E(\mathbb{F}_p)|$. This proves that M and E have the same order.

2. $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

Because the curve M'' has a point of order two at $(0, 0)$, the $\mathbb{F}_p$ -isomorphic curve E also has a point of order two. By Lemma 2, we denote it as $(r, 0)$, and therefore the equation $z^3 + az + b = 0$ has at least one root. Furthermore, the discriminant of this equation is

$$D = -(4a^3 + 27b^2) = -\left(-\frac{16(A^2 - 4)^2}{B^6}\right) = \frac{16(A^2 - 4)^2}{B^6}.$$

This is a non-zero quadratic residue (note: $B(A^2 - 4) \neq 0$), so by Lemma 5, it follows that $z^3 + az + b = 0$ has three distinct roots. Therefore, $E(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. □

Theorem 5. Let $E_{a,b}$ denote an elliptic curve over $\mathbb{F}_p$ given by the equation $y^2 = x^3 + ax + b$, and let $M_{A,B}$ denote a Montgomery curve given by $By^2 = x^3 + Ax^2 + x$. For a prime $p > 3$ and a trace of Frobenius t , we define two sets:

1. $S_E(p, t) = \{(a, b) \in \mathbb{F}_p^2 \mid |E_{a,b}(\mathbb{F}_p)| = p + 1 - t, E_{a,b}(\mathbb{F}_p)[2] \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \text{ and } 4a^3 + 27b^2 \neq 0\},$
2. $S_M(p, t) = \{(A, B) \in \mathbb{F}_p^2 \mid |M_{A,B}(\mathbb{F}_p)| = p + 1 - t \text{ and } B(A^2 - 4) \neq 0\}.$

For any prime $p > 3$ and t , $|S_M(p, t)| = 6|S_E(p, t)|$.

Proof. 1. When $4 \nmid p + 1 - t$, the two sets are empty. When $t^2 > 4p$, the sets are empty. In these cases, $|S_M(p, t)| = 6|S_E(p, t)| = 0$.

2. We now consider the case where $t^2 \leq 4p$ and $4 \mid (p + 1 - t)$.

Let us define a map f that takes an element $(a, b) \in S_E(p, t)$ and maps it to the set of six Montgomery coefficient pairs derived from it, as per Lemma 13.

$$f((a, b)) = \{(\pm A_1, \pm B_1), (\pm A_2, \pm B_2), (\pm A_3, \pm B_3)\}.$$

The map f is well-defined for every element of $S_E(p, t)$ because membership in this set guarantees the existence of the three distinct roots and the non-zero quadratic residues required to construct the coefficients.

From our preceding lemmas, the map f has the following properties:

- The six pairs in $f((a, b))$ are mutually distinct, so $|f((a, b))| = 6$.
- Each pair in $f((a, b))$ defines a non-singular Montgomery curve with order $p + 1 - t$, which means $f((a, b)) \subset S_M(p, t)$.
- By Lemma 12, for any two distinct pairs $(a, b), (a', b') \in S_E(p, t)$, their images are disjoint: $f((a, b)) \cap f((a', b')) = \emptyset$.

The set $S_M(p, t)$ contains the union of all these disjoint image sets. Therefore,

$$|S_M(p, t)| \geq \left| \bigcup_{(a,b) \in S_E(p,t)} f((a,b)) \right| = \sum_{(a,b) \in S_E(p,t)} |f((a,b))| = 6|S_E(p, t)|.$$

Let us define a map $g : S_M(p, t) \rightarrow S_E(p, t)$ using the explicit formulas:

$$g((A, B)) = (a, b) = \left(-\frac{A^2 + 12}{3B^2}, \frac{2A(A^2 - 36)}{27B^3} \right).$$

This map is well-defined for all $(A, B) \in S_M(p, t)$ since the non-singularity condition guarantees $B \neq 0$.

Then, we analyze the size of the preimage set $g^{-1}((a, b))$ for any (a, b) in the image of g . This requires finding the number of solutions (A, B) to the system of equations:

$$\begin{aligned} a &= -\frac{A^2 + 12}{3B^2}, \\ b &= \frac{2A(A^2 - 36)}{27B^3}. \end{aligned}$$

Thus,

$$B^6 = (B^2)^3 = \left(-\frac{A^2 + 12}{3a}\right)^3 = -\frac{(A^2 + 12)^3}{27a^3},$$

$$B^6 = (B^3)^2 = \left(\frac{2A(A^2 - 36)}{27b}\right)^2 = \frac{4A^2(A^2 - 36)^2}{729b^2}.$$

Equating these two expressions for B^6 yields a single equation involving only A . This equation is a 6th-degree polynomial in A , and more specifically, a cubic polynomial in A^2 . A cubic equation has at most 3 roots for A^2 . Each solution for A^2 yields at most 2 roots for A . Thus, there are at most 6 possible values for A . For each such solution for A , the value of B is uniquely determined by the relation $B = B^3/B^2 = -\frac{2aA(A^2-36)}{9b(A^2+12)}$.

This shows that the preimage has a size of at most 6: $|g^{-1}((a, b))| \leq 6$. Therefore,

$$|S_M(p, t)| = \left| \bigcup_{(a, b) \in \text{Image}(g)} g^{-1}((a, b)) \right| \leq \sum_{(a, b) \in \text{Image}(g)} |g^{-1}((a, b))| \leq 6|\text{Image}(g)|.$$

Since the image is a subset of the $S_E(p, t)$, $|\text{Image}(g)| \leq |S_E(p, t)|$. Combining these, we get:

$$|S_M(p, t)| \leq 6|S_E(p, t)|.$$

Since we have established both $|S_M(p, t)| \geq 6|S_E(p, t)|$ and $|S_M(p, t)| \leq 6|S_E(p, t)|$, we conclude that:

$$|S_M(p, t)| = 6|S_E(p, t)|.$$

□

Lemma 15. Let $p > 3$ be a prime. The cardinality of the set $S_E(p, t)$, as defined in Theorem 5, is given by the formula:

$$|S_E(p, t)| = \begin{cases} \frac{p-1}{2} H_w \left(\frac{t^2 - 4p}{4} \right) & \text{if } 4 \mid p + 1 - t \text{ and } t^2 < 4p, \\ 0 & \text{otherwise} \end{cases},$$

where H_w is the Kronecker-Hurwitz class number.

Proof. See Lemma 3 in [4]. Note that for any prime $p > 3$, $h_w(-p) = H_w(-p)$. □

Theorem 6. Let $p > 3$ be a prime. The cardinality of the set $S_M(p, t)$, as defined in Theorem 5, is given by the following formula:

$$|S_M(p, t)| = \begin{cases} 3(p-1) H_w \left(\frac{t^2 - 4p}{4} \right) & \text{if } 4 \mid p + 1 - t \text{ and } t^2 < 4p, \\ 0 & \text{otherwise} \end{cases},$$

where H_w is the Kronecker-Hurwitz class number.

Proof. This result is obtained by combining Theorem 5 and Lemma 15. □

Acknowledgements

I would like to acknowledge the assistance of DeepL and Gemini in translating and refining the English text of this paper. All mathematical ideas, including the theorems, propositions, and their original proofs, were my own work, initially completed in my native language.

References

- [1] Daniel J. Bernstein. “Curve25519: New Diffie-Hellman Speed Records”. In: *Public Key Cryptography - PKC 2006*. Ed. by Moti Yung et al. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006, pp. 207–228. ISBN: 9783540338529.
- [2] B. J. Birch. “How the Number of Points of An Elliptic Curve Over a Fixed Prime Field Varies”. In: *Journal of the London Mathematical Society* s1-43.1 (1968), pp. 57–60. DOI: <https://doi.org/10.1112/jlms/s1-43.1.57>. eprint: <https://londmathsoc.onlinelibrary.wiley.com/doi/pdf/10.1112/jlms/s1-43.1.57>. URL: <https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/jlms/s1-43.1.57>.
- [3] Steven D. Galbraith and James McKee. “The Probability that the Number of Points on an Elliptic Curve over a Finite Field is Prime”. In: *Journal of the London Mathematical Society* 62.3 (2000), pp. 671–684. DOI: <https://doi.org/10.1112/S0024610700001502>. eprint: <https://londmathsoc.onlinelibrary.wiley.com/doi/pdf/10.1112/S0024610700001502>. URL: <https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/S0024610700001502>.
- [4] Nathan Kaplan and Ian Petrow. “Elliptic curves over a finite field and the trace formula”. In: *Proceedings of the London Mathematical Society* 115.6 (2017), pp. 1317–1372.
- [5] James McKee. “Subtleties in the Distribution of the Numbers of Points on Elliptic Curves Over a Finite Prime Field”. In: *Journal of the London Mathematical Society* 59.2 (1999), pp. 448–460. DOI: <https://doi.org/10.1112/S0024610799007334>. eprint: <https://londmathsoc.onlinelibrary.wiley.com/doi/pdf/10.1112/S0024610799007334>. URL: <https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/S0024610799007334>.
- [6] Katsuyuki Okeya, Hiroyuki Kurumatani, and Kouichi Sakurai. “Elliptic Curves with the Montgomery-Form and Their Cryptographic Applications”. In: *Public Key Cryptography*. Ed. by Hideki Imai and Yuliang Zheng. Berlin, Heidelberg: Springer Berlin Heidelberg, 2000, pp. 238–257. ISBN: 9783540465881.
- [7] René Schoof. “Nonsingular plane cubic curves over finite fields”. In: *Journal of Combinatorial Theory, Series A* 46.2 (1987), pp. 183–211. ISSN: 0097-3165.

- [8] Joseph H. Silverman. *The Arithmetic of Elliptic Curves*. 2nd ed. Springer New York, NY, 2009. ISBN: 9780387094946.
- [9] Douglas R. Stinson. *Cryptography: Theory and Practice*. 3rd ed. Chapman & Hall/CRC, 2005. ISBN: 9781420057133.
- [10] Zhi-Hong Sun. “Cubic and quartic congruences modulo a prime”. In: *Journal of Number Theory* 102.1 (2003), pp. 41–89. ISSN: 0022-314X.
- [11] Lawrence C. Washington. *Elliptic Curves: Number Theory and Cryptography*. 2nd ed. Chapman & Hall/CRC, 2008. ISBN: 9781420071467.